

SMOKY LAKE AGDM, AB, Canada

WMO: 712740

Lat: 54.265N

Lon: 112.504W

Elev: 2231

StdP: 13.55

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-25.8	-19.8	-32.1	1.0	-25.3	-25.7	1.4	-19.3	23.0	28.4	20.2	26.2	5.0	140	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.7	81.8	63.3	78.4	62.0	75.3	60.4	65.8	77.8	63.8	75.5	61.9	72.4	8.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	87.7	71.4	59.3	81.7	68.7	57.5	76.7	66.8	31.8	77.7	30.2	75.8	28.8	72.5	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	17.9	15.7	DB	-36.2	87.6	7.3	3.6	-41.4	90.2	-45.7	92.3	-49.7	94.3	-55.0	96.9	(4)
(5)			WB	-37.2	70.4	7.3	2.0	-42.4	71.8	-46.7	73.0	-50.8	74.1	-56.1	75.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.8	10.1	10.7	23.2	37.8	50.1	56.6	61.4	58.8	49.8	37.9	21.3	9.7		
	DBStd	21.83	15.94	13.91	12.73	9.41	8.07	5.56	5.45	6.25	7.97	8.74	12.42	14.19		
	HDD50	6254	1237	1102	830	377	99	11	0	6	99	384	860	1249		
	HDD65	10726	1702	1522	1295	817	463	256	140	210	457	840	1310	1714		
	CDD50	1056	0	0	0	10	103	209	353	278	94	9	0	0		
	CDD65	53	0	0	0	0	2	5	28	16	2	0	0	0		
	CDH74	956	0	0	0	1	72	113	420	262	86	2	0	0		
	CDH80	180	0	0	0	0	9	11	98	46	16	0	0	0		
	WSAvg	7.1	6.8	6.8	7.8	8.6	8.2	7.2	6.5	6.4	6.8	7.3	6.7	6.1		
	PrecAvg	15.9	0.7	0.5	0.6	0.9	1.6	2.8	3.1	2.0	1.3	0.7	0.6	0.7		
(15) Precipitation	PrecMax	20.0	1.5	1.1	1.3	2.2	3.4	5.4	6.1	3.5	2.9	2.2	1.6	1.5		
	PrecMin	11.8	0.0	0.0	0.1	0.1	0.2	0.9	0.9	0.8	0.4	0.2	0.2	0.0		
	PrecStd	2.4	0.4	0.3	0.4	0.5	0.7	1.2	1.1	0.7	0.7	0.4	0.4	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.0	44.5	52.7	71.1	81.0	81.2	87.5	84.9	82.5	71.2	52.2	42.8		
		MCWB	37.2	36.5	42.2	50.7	56.9	61.8	66.8	64.4	60.8	53.4	44.1	36.1		
	2%	DB	39.3	39.3	46.6	64.4	75.3	77.3	82.6	80.2	75.7	61.2	44.7	37.6		
		MCWB	34.6	33.8	38.3	46.6	54.2	60.2	65.3	63.0	58.8	49.1	38.7	32.9		
	5%	DB	35.7	35.3	42.9	58.6	70.8	73.7	78.6	76.6	70.1	55.8	40.2	33.2		
		MCWB	32.1	31.1	36.1	44.0	52.4	58.0	63.9	62.0	55.5	45.5	35.3	29.9		
	10%	DB	31.8	31.2	39.5	53.7	66.8	70.1	75.1	73.0	64.7	51.5	36.1	29.1		
		MCWB	29.3	28.1	33.8	41.5	50.6	56.5	62.1	60.6	53.6	42.9	32.5	26.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.8	37.3	42.9	51.2	58.3	64.7	69.6	67.8	62.4	54.1	43.9	36.8		
		MCDB	42.9	43.2	50.5	70.2	74.0	77.6	80.7	79.4	78.9	68.8	51.3	42.0		
	2%	WB	35.1	34.0	39.1	47.4	55.9	62.1	66.7	65.1	59.4	49.4	39.3	33.3		
		MCDB	38.8	38.7	46.2	62.7	71.3	72.9	78.9	76.8	73.1	58.7	43.8	37.4		
	5%	WB	32.4	31.2	36.6	44.6	53.9	60.0	64.8	63.0	56.9	46.0	35.7	30.2		
		MCDB	35.4	34.9	42.2	57.3	67.5	70.8	76.8	74.1	69.1	53.9	39.7	33.3		
	10%	WB	29.2	27.8	34.3	42.2	51.7	58.0	62.9	61.1	54.5	43.6	32.4	27.1		
		MCDB	31.7	31.0	39.1	52.4	64.3	67.8	73.5	71.4	63.6	50.1	35.2	29.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.6	20.2	18.1	20.3	23.4	21.8	22.7	22.9	22.4	18.2	14.9	16.1		
		MCDBR	19.4	24.7	20.8	28.9	29.3	28.0	28.8	28.9	32.4	26.6	18.6	18.8		
		MCWBR	15.5	19.9	13.8	15.2	13.6	13.5	14.1	14.0	16.5	15.5	12.8	15.4		
		MCDBR	19.4	24.2	19.8	27.4	26.8	24.0	26.4	26.2	29.0	24.1	17.1	19.0		
	5% WB	MCWBR	15.9	19.5	13.5	15.2	13.0	12.6	13.7	13.7	15.1	14.8	12.5	15.9		
(40) Clear-Sky Solar Irradiance	taub		0.250	0.268	0.301	0.327	0.339	0.352	0.352	0.361	0.307	0.278	0.248	0.236		
	taud		2.283	2.303	2.315	2.380	2.401	2.379	2.426	2.351	2.511	2.561	2.454	2.279		
	Ebn at Noon		230	266	279	286	288	284	282	271	277	264	236	211		
	Edh at Noon		18	24	31	33	35	36	34	34	25	19	15	14		
(44) All-Sky Solar Radiation	RadAvg		275	600	1080	1477	1802	1886	1909	1566	1080	608	284	200		
	RadStd		31	43	81	124	147	126	101	111	111	80	31	17		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+299	N/A
Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page